Hongxuan Chen

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https://hxchen9.github.io/

Education

University of Illinois at Urbana-Champaign
Ph.D. in Computer Science (Expected Spring 2026)
- Advisor: Dr. Geoffrey Herman & Dr. Jeff Erickson
- GPA: 4.0

University of Illinois at Urbana-Champaign
B.S. in Computer Science, *Highest Honors* (May 2021)
- GPA: 4.0

Research Interests

My research focuses on computer science education, particularly:
- How students learn algorithm design and discrete mathematics
- Collaborative learning in CS classrooms
- Curriculum evaluation and education policies

Teaching Experience

Co-Instructor
CS173: Discrete Structures – University of Illinois Urbana-Champaign, Summer 2025
- Co-taught with another Ph.D. student without faculty supervision
- Designed and delivered lectures and discussion sessions; provided student support through office hours; managed course staff of 8 and logistics for a class of ~50 students

CS233: Computer Architecture – University of Illinois Urbana-Champaign, Spring 2024
- Co-taught with another Ph.D. student under faculty supervision
- Delivered lectures and led collaborative learning sessions; managed course logistics for ~280 students and 35 staff members

Graduate Teaching Assistant
CS233: Computer Architecture – University of Illinois Urbana-Champaign, Fall 2021
CS173: Discrete Structures – University of Illinois Urbana-Champaign, Summer 2022
- Led discussion sections, held office hours, and organized review sessions
- Developed supplementary course materials and provided student support

Undergraduate Course Assistant
CS125, CS233, CS374 – University of Illinois Urbana-Champaign, Spring 2018 - Spring 2021
- Conducted quality assurance for course materials
- Assisted students through office hours, discussion sessions, and review sessions
- Graded assignments and provided detailed feedback

Peer-reviewed Publications

- Gertner, Y., Solomon, B., **Chen, H.**, Robson, E., Evans, C., Erickson, J. (2026). Measuring Students' Perceptions of an Autograded Scaffolding Tool for Students Performing at All Levels in an Algorithms Class. Accepted by *the 57th ACM Technical Symposium on Computer Science Education.*

- Liu, J., Poulsen, S., Goodwin, E., **Chen, H.**, Williams, G., Gertner, Y., & Franklin, D. (2025). Teaching algorithm design: a literature review. *ACM Transactions on Computing Education*, *25*(2), 1-20. https://doi.org/10.1145/37279

- (Poster) Liu, J., Poulsen, S., **Chen, H.**, Williams, G., Gertner, Y., & Franklin, D. (2024). Teaching Algorithm Design: A Literature Review. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2* (pp. 1722-1723). https://doi.org/10.1145/3626253.3635548

- (Poster) **Chen, H.**, Braught, K., Herman, G. L., & Erickson, J. (2025). Novice Difficulties in Graph Layering for Algorithm Design. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 2* (pp. 1415-1416). https://doi.org/10.1145/3641555.3705221

- Poulsen, S., **Chen, H.**, Gertner, Y., Cosman, B., West, M., & Herman, G. L. (2025). Measuring the Impact of Distractors on Student Learning Gains while Using Proof Blocks. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (pp. 917-923). https://doi.org/10.1145/3641554.3701787

- **Chen, H.**, Li, A., Challen, G., & Cunningham, K. (2024). Implementation of split deadlines in a large cs1 course. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (pp. 193-199). https://doi.org/10.1145/3626252.3630873

- Poulsen, S., Gertner, Y., **Chen, H.**, Cosman, B., West, M., & Herman, G. L. (2024). Disentangling the learning gains from reading a book chapter and completing proof blocks problems. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (pp. 1056-1062). https://doi.org/10.1145/3626252.3630831

- **Chen, H.**, Fong, M. M., Herman, G. L., & Silva, M. (2023). Student Autonomy in Collaborative Learning: Effects of meeting time and team consistency. In *2023 IEEE Frontiers in Education Conference (FIE)* (pp. 1-9). IEEE. https://doi.org/10.1109/FIE58773.2023.10343523

- **Chen, H.**, West, M., Hilgenfeldt, S., & Silva, M. (2023). Measuring the Impact of a Computational Linear Algebra Course on Students' Exam Performance in a Subsequent Numerical Methods Course. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (pp. 115-121). https://doi.org/10.1145/3545945.3569778

- Fong, M. M., Butler, L., **Chen, H.**, & Herman, G. L. (2022). Validating an Observation Protocol for Structured Roles in Cooperative Learning. In *2022 IEEE Frontiers in Education Conference (FIE)* (pp. 1-9). IEEE. https://doi.org/10.1109/FIE56618.2022.9962529

- Fong, M., **Chen, H.**, Butler, L., & Herman, G. (2022, August). Developing an Observation Protocol for Cooperative Learning. In *2022 ASEE Annual Conference & Exposition*. https://doi.org/10.18260/1-2--40991

- Mahmood, M. S., **Chen, H.**, Fong, M., & Herman, G. (2022, August). Work in Progress: Exploring Students' Misconceptions of Cache Memories. In *2022 ASEE Annual Conference & Exposition*. https://doi.org/10.18260/1-2--40679

Invited Talks

- *Implementation of Split Deadlines in a Large CS1 Course,* presented remotely with Geoffrey Challen at the Computing Education Seminar, **UC Davis**, Oct. 2024

Service & Outreach

Reviewer

- ACM SIGCSE TS 2023, 2025
- IEEE FIE 2025

Student Volunteer

- ACM SIGCSE TS 2024, 2025

Teaching Assistant (Girls Who Code, Remote, Summer 2020)

- Assisted instructors during online lectures, grading, and office hours
- Organized advisory panels to encourage female high school students to pursue CS

Honors and Awards

- **Yee Memorial Fund Fellowship** ($5,000), Grainger College of Engineering, UIUC, 2025-2026
- **Teacher Ranked as Excellent by Students**, UIUC, Summer 2022, Spring 2024, Summer 2025
- **Outstanding Course Assistant**, CS Department, UIUC (5 recipients selected), Spring 2021
- **Bronze Tablet Honoree**, UIUC (Top 3% of graduating class), Spring 2021
- **Illinois Engineering Achievement Scholarship** ($1,000), Spring 2021

Industry Experience

Data Scientist Intern (Double Dragon Big Data, Guiyang, China, Summer 2019)

- Developed web scrapers to extract data from Baidu Baike
- Designed algorithms for data cleaning and network analysis of word relationships

Technical Skills

- Programming: Java, Python, C, JavaScript, TypeScript
- App Development: React Native (TypeScript, JavaScript)
- Web Development: HTML/CSS/JavaScript, React
- Database Management: SQL, MongoDB, Neo4j
- Data Analysis: Python, R

Languages

- Mandarin Chinese: Native
- English: Fluent (TOEFL iBT: 117/120, July 2020)